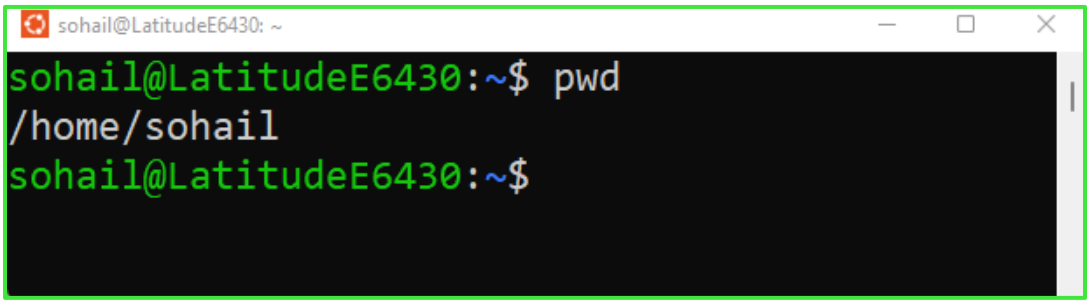
ASSIGNMENT NO. O1	SUBMITTED BY SOHAIL ASLAM BHATTI
# BASIC LINUX COMMANDS	SUBMITTED TO SIR TAIMOOR KAMRAN
	DEVOPS BATCH NO. 2
	ROLL NO. 912687

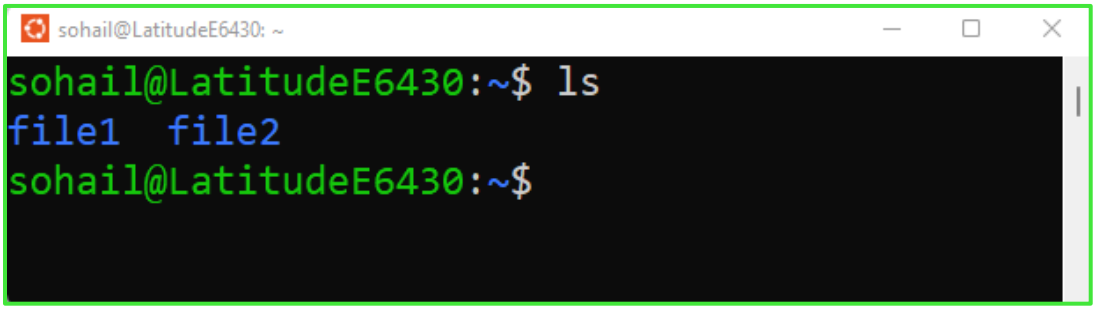
1. Use the pwd command to display your current working directory.

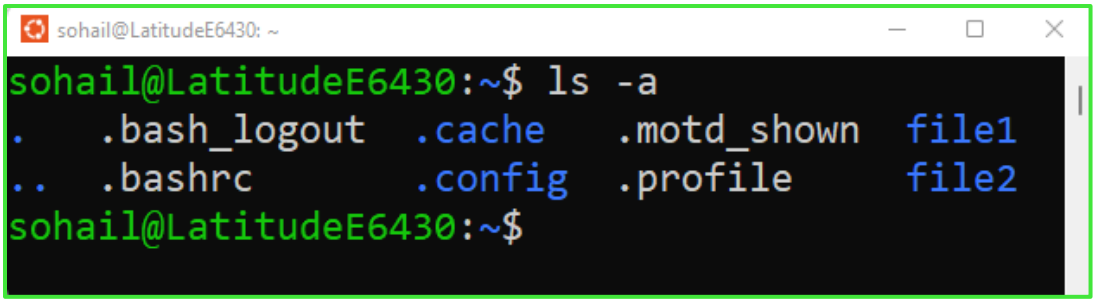


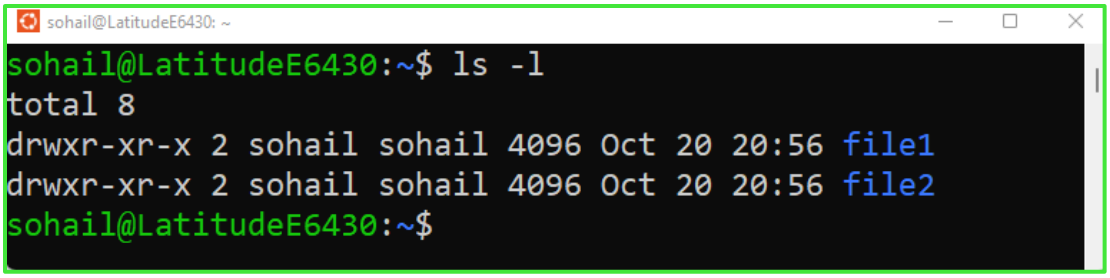
Explanation:

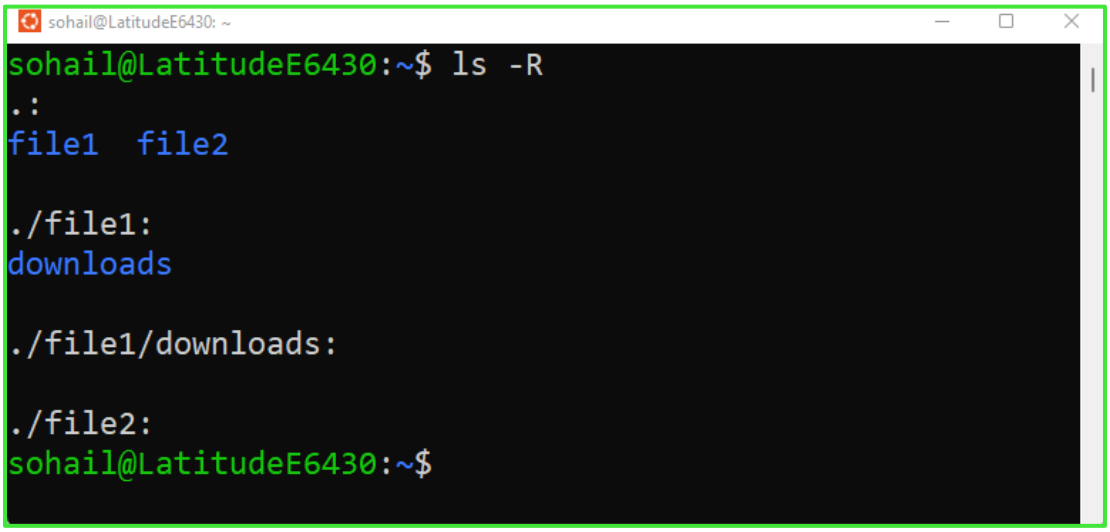
The **pwd** (Print Working Directory) command displays the full path of the directory in which you are currently working.

2. Use the ls, ls -a, ls -l, and ls -R commands to list files and directories in different ways. Share the outputs and explain the difference between these commands.





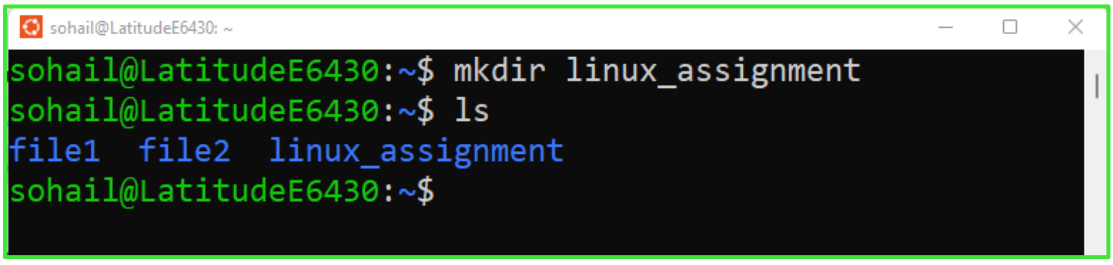




Explanation:

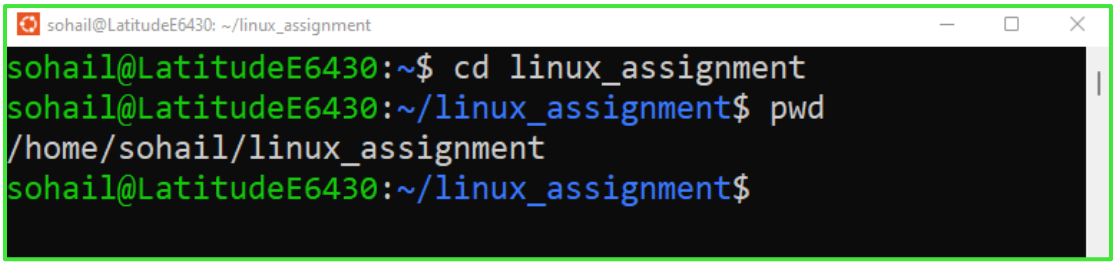
Command	Purpose
ls	Lists files and directories.
ls -a	Lists all entries, including hidden files.
ls -l	Displays detailed information such as permissions, owner, size, and modification date.
ls -R	Lists directories and their contents recursively, including subdirectories.

3. **Use the mkdir command to create a directory named linux_assignment. Then use ls to verify that it was created.**



Explanation: The **mkdir** command creates a new directory. The **ls** command verifies that the directory exists.

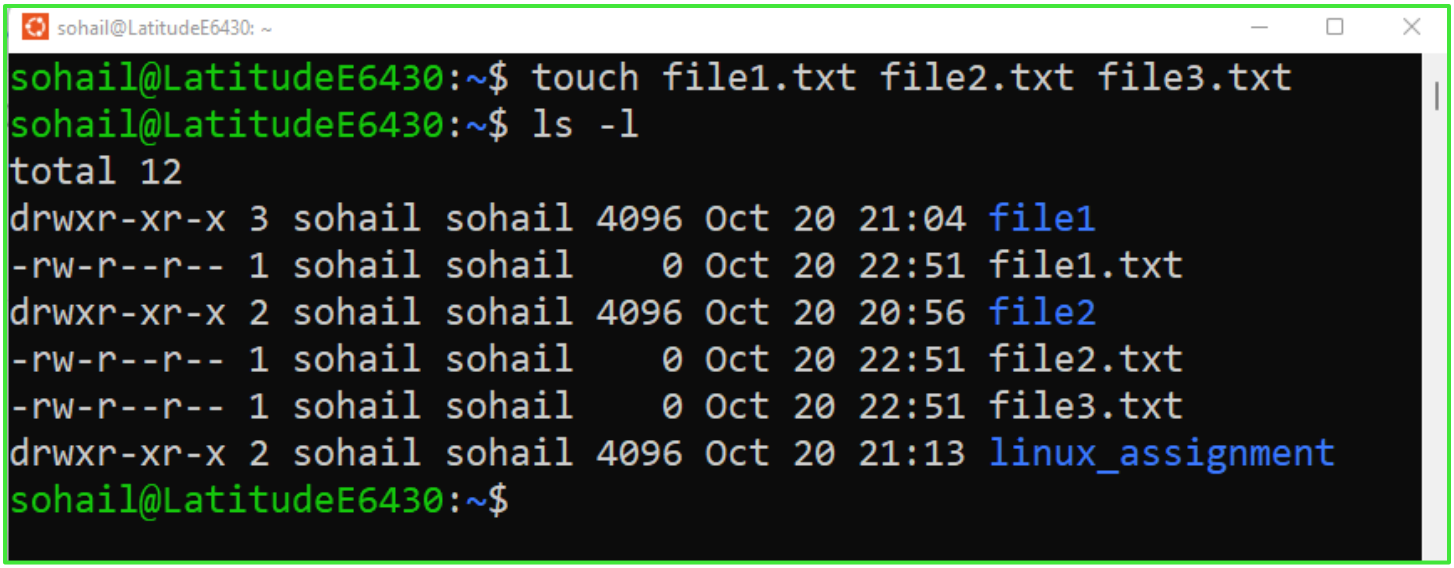
4. **Use the cd command to enter the linux_assignment directory. Use pwd to confirm your current location.**



Explanation:

The **cd** command changes the current directory, and **pwd** confirms the new location.

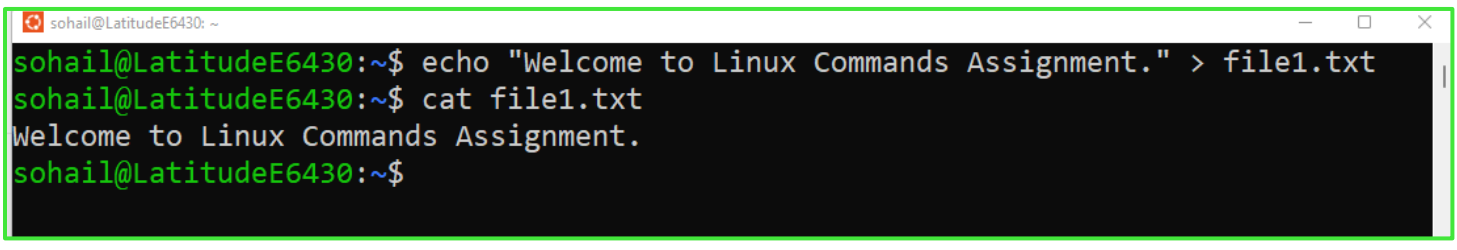
5. Create three empty files named file1.txt, file2.txt, and file3.txt using the touch command. Use ls -l to verify their creation.



Explanation:

The **touch** command creates empty files if they do not exist. The **ls -l** command displays their details. A size of 0 indicates an empty file.

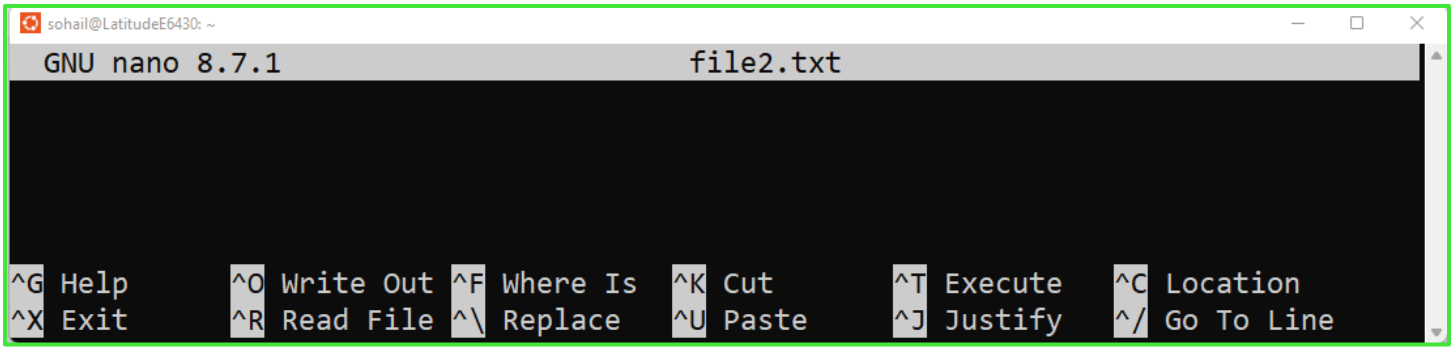
6. **Use the echo command to write a message into file1.txt. Then use the cat command to display the contents of the file.**

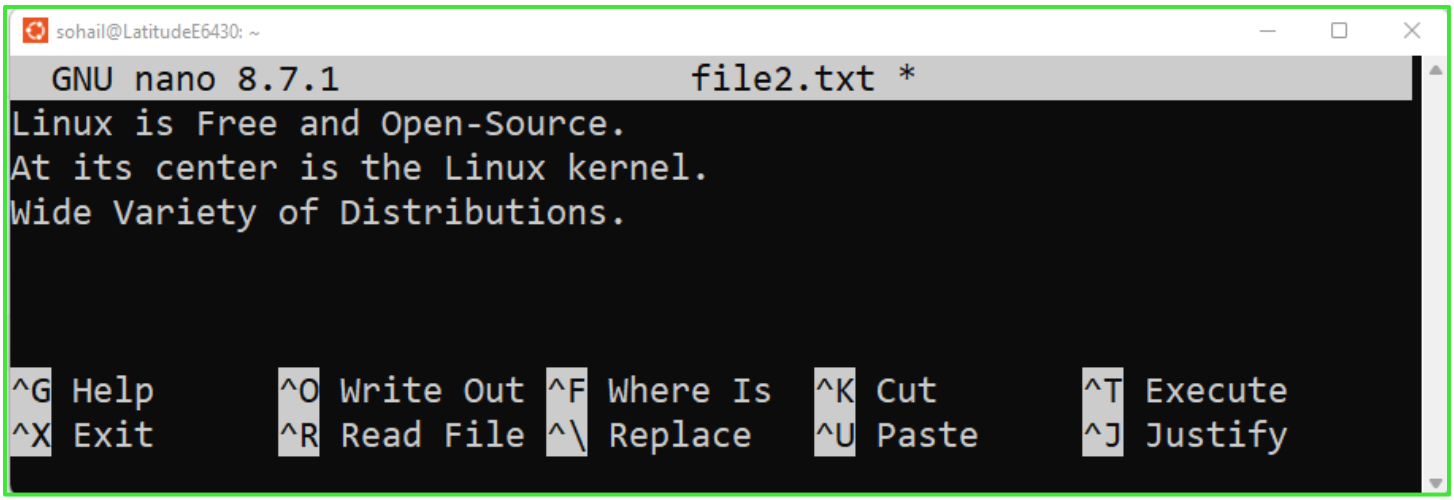


Explanation:

- **echo** displays text or sends it to a file.
- **>** redirects the text into file1.txt, replacing its existing contents.
- **cat** displays the contents of the file.

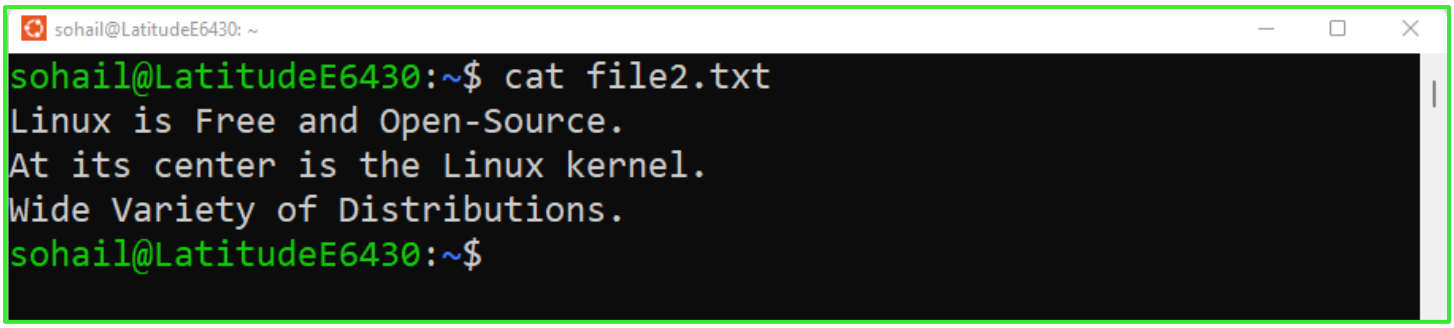
7. **Use the nano command to create or edit file2.txt. Write at least three lines of text, save the file, and display its contents using cat.**





Save and exit Nano:

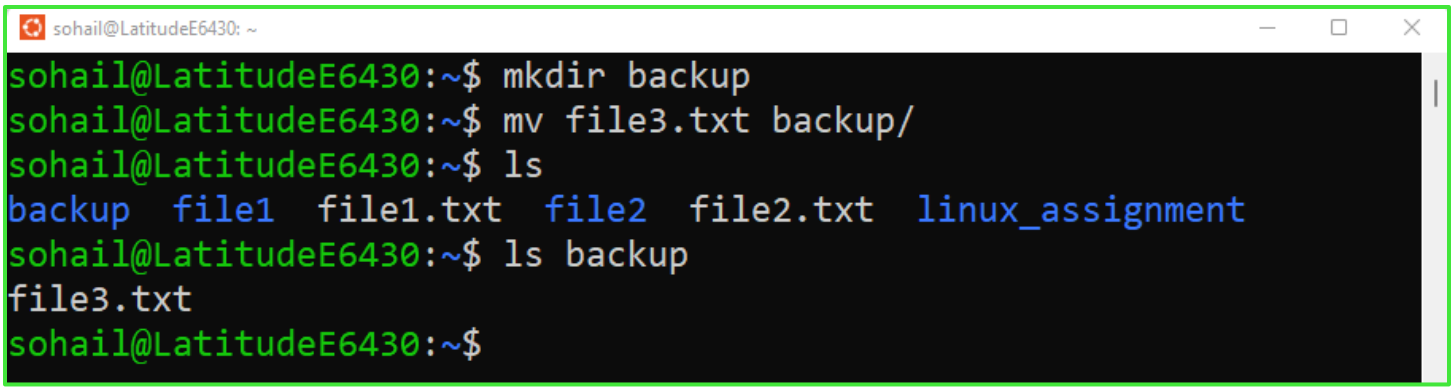
1. Press **Ctrl** + **O** to save the file.
2. Press **Enter** to confirm the filename.
3. Press **Ctrl** + **X** to exit.



Explanation:

The **nano** command opens a simple text editor that allows users to create and edit files through the terminal.

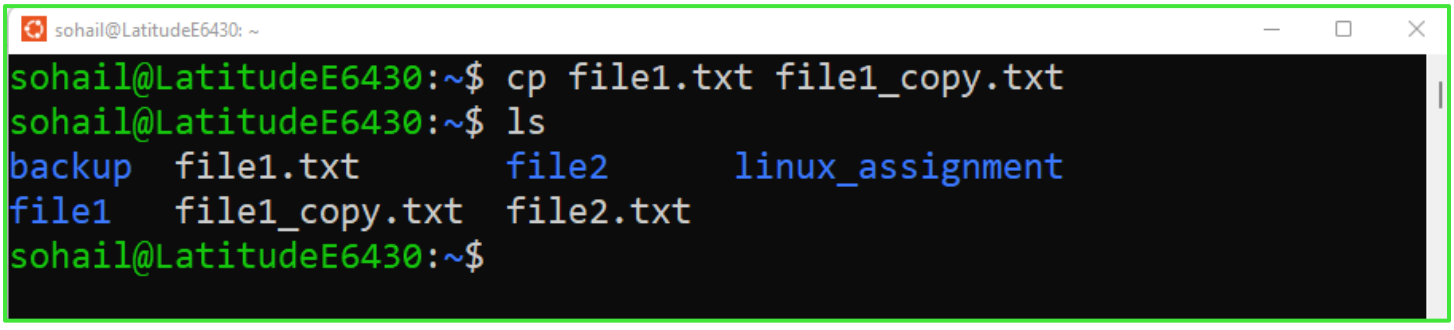
8. **Create a directory named backup inside linux_assignment. Use the mv command to move file3.txt into the backup directory. Verify the result using ls.**



Explanation:

The **mv** command moves a file from one location to another. After moving the file, **file3.txt** is no longer in the main directory; it is inside backup.

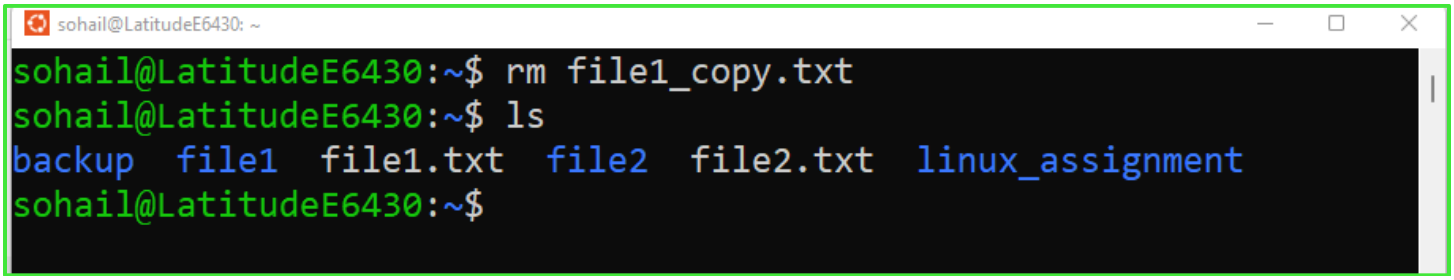
9. **Use the cp command only if you have covered it in class; otherwise, skip this question. Copy file1.txt to a new file named file1_copy.txt and verify that both files exist.**



Explanation:

The **cp** command copies a file. The original **file1.txt** remains unchanged, and a new file named **file1_copy.txt** is created.

10. **Use the rm command to delete file1_copy.txt. Use ls to confirm that the file has been removed.**



Explanation:

The **rm** command removes files. The output confirms that **file1_copy.txt** is no longer listed.

11. **Use the cd .., cd ~, and cd / commands one by one. After each command, use pwd to show which directory you are in.**

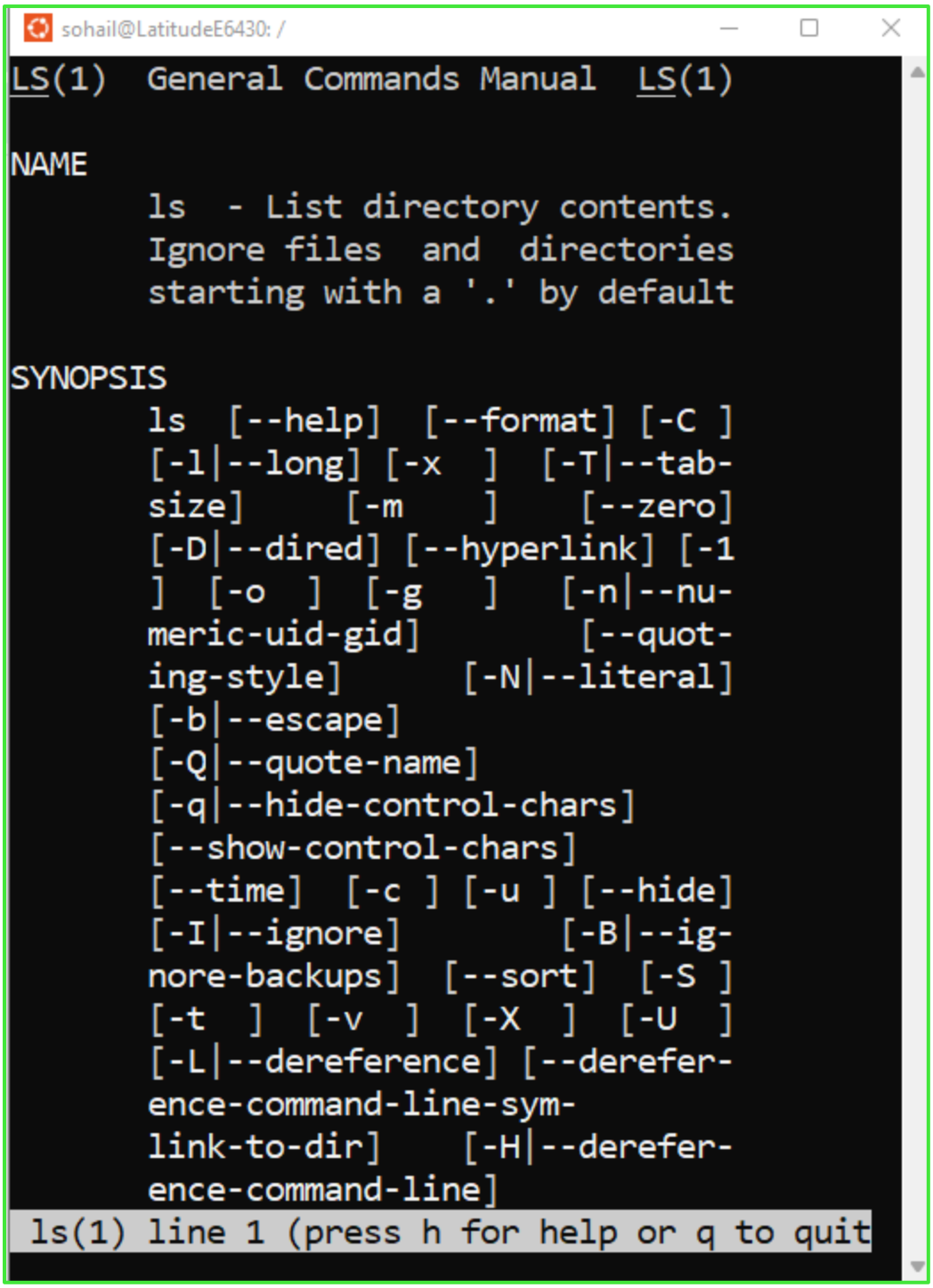


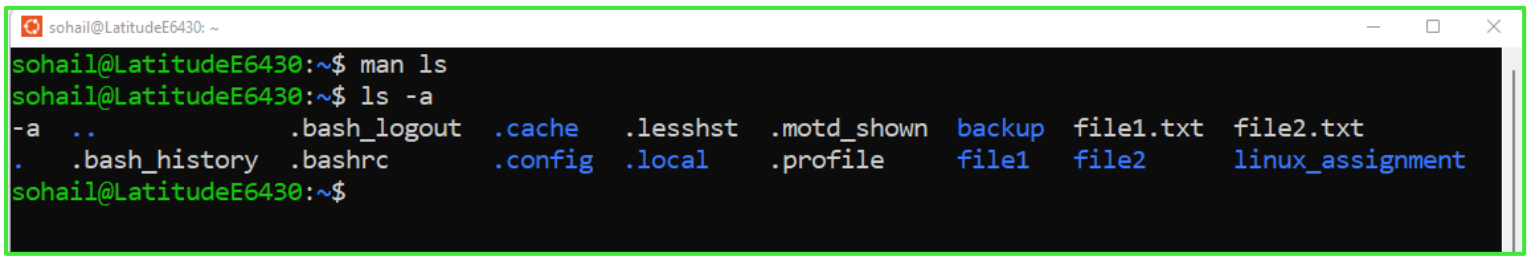
Explanation:

Command	Purpose
cd ..	Moves to the parent directory.
cd ~	Moves to the current user's home directory.
cd /	Moves to the root directory of the filesystem.

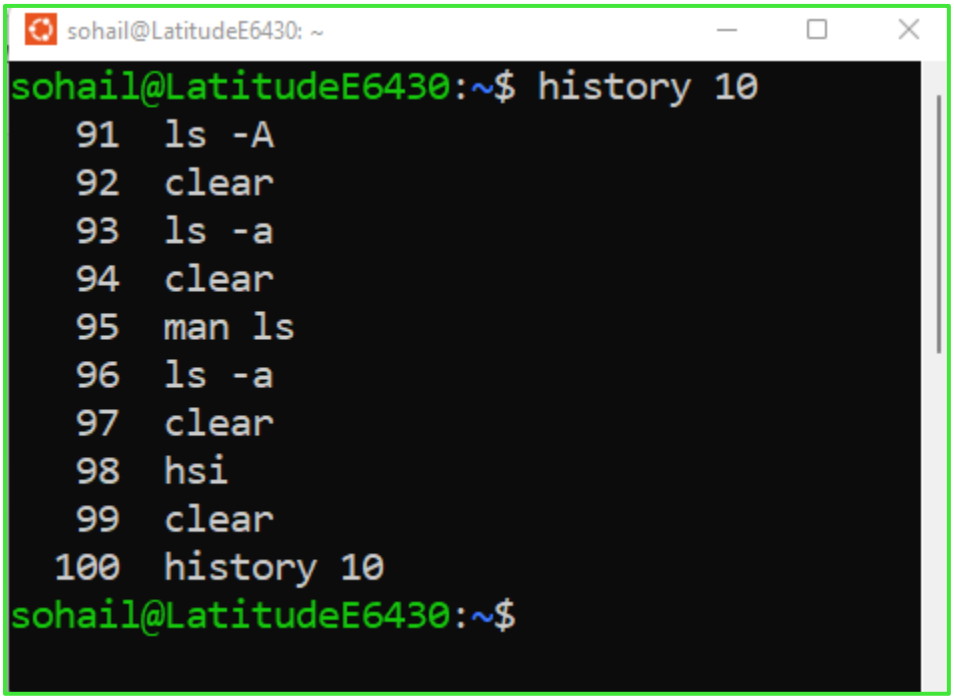
The output of **cd ..** depends on your location before running it.

12. Use the man command to open the manual for ls. Find the description of the -a option. Share what it does.





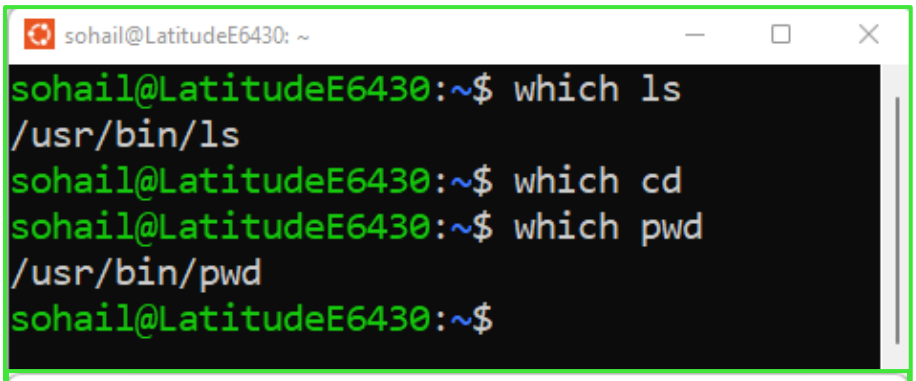
13. Use the history command to display your previously executed commands. Share at least 10 commands from your terminal history.

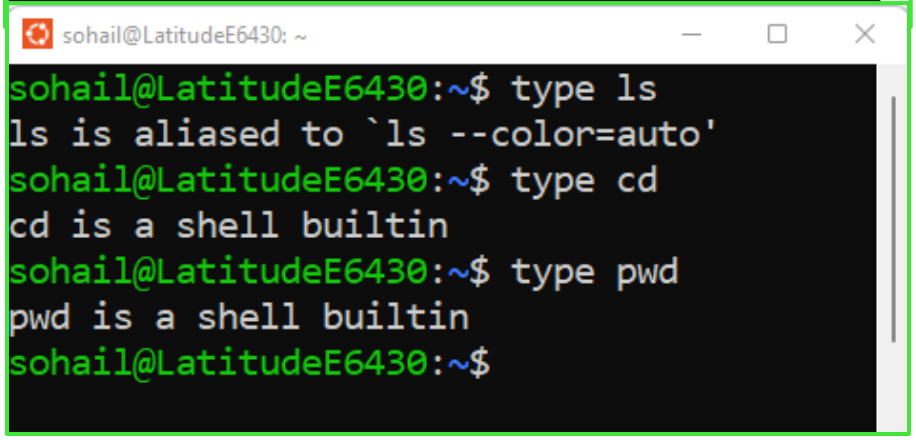


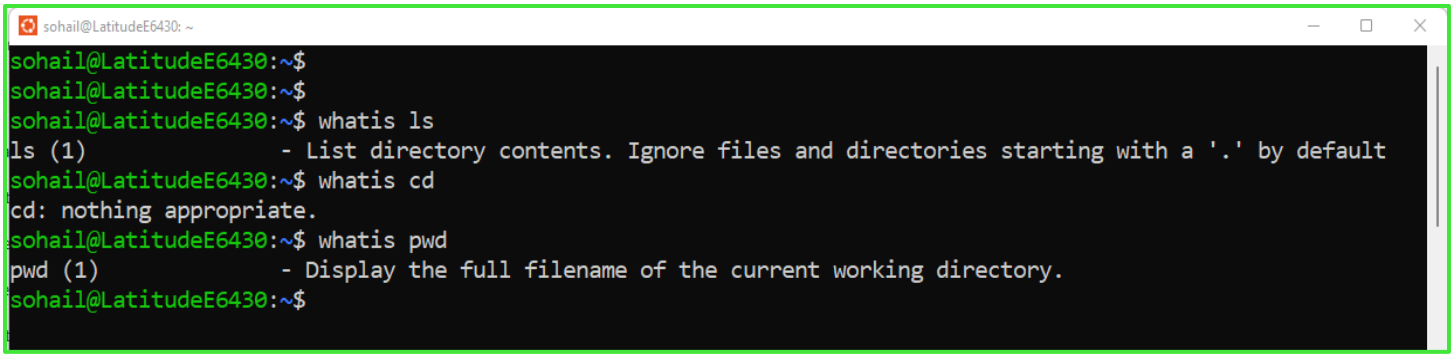
Explanation:

The **history** command displays previously executed commands in the shell.

14. Use the which, type, and whatis commands on ls, cd, and pwd. Record the outputs and identify which commands are shell built-ins and which are external programs, where applicable.







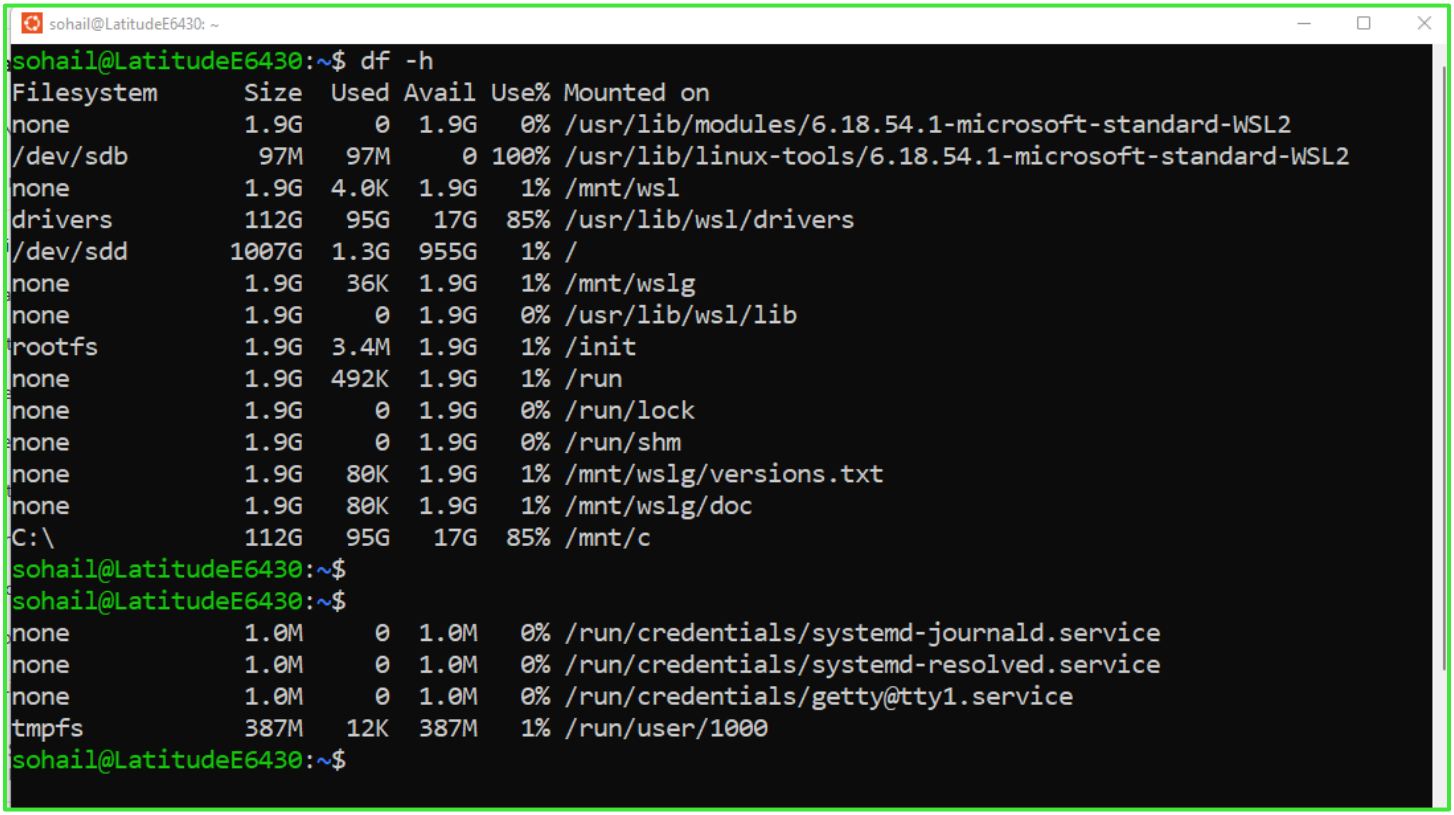
The **whatis** output may vary depending on installed manual pages and database configuration.

Which commands are built-ins or external programs?

Command	Classification	Explanation
ls	External program	Usually available as /usr/bin/ls.
cd	Shell built-in	Changes the shell's current directory.
pwd	Both commonly available	Often a shell built-in, with an external executable also available.

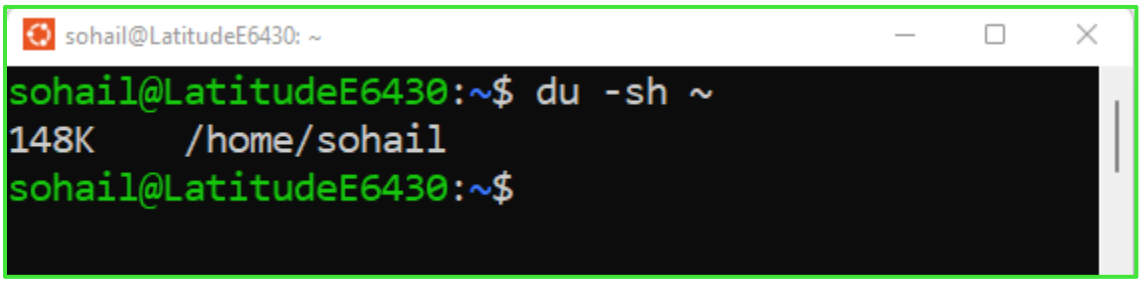
Conclusion: **type** is particularly useful for identifying whether a command is a **shell built-in** or an **external program.**

15. **Use the df and du commands to check disk usage. Run df -h to view file system space and du -sh ~ to check the total size of your home directory. Share both outputs.**



Explanation: The **df -h** command displays filesystem disk usage in human-readable units, such as KB, MB, and GB.

- **Filesystem:** The storage device or filesystem.
- **Size:** Total filesystem capacity.
- **Used:** Space already occupied.
- **Avail:** Available space.
- **Use%:** Percentage of space used.
- **Mounted on:** The location where the filesystem is accessible.



Explanation:

- **du**: Estimates disk space used by files and directories.
- **-s**: Displays a summary instead of listing every subdirectory.
- **-h**: Displays sizes in human-readable units.
- **~**: Represents the current user's home directory.

The output indicates that the home directory occupies approximately 148K.